



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (INFORMATION SHEET)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-1:2015CORE ABILITIES	
TAHAP / LEVEL	1 (ONE)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA04 HEALTH, SAFETY AND ENVIRONMENTAL AWARENESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	4.1 ADHERE TO HEALTH AWARENESS ACTIVITY 4.2 ADHERE TO SAFETY AWARENESS ACTIVITY 4.3 ADHERE TO ENVIRONMENTAL AWARENESS ACTIVITY	
NO. KOD / CODE NO.	Z-009-1:2015–M04/P(3/5)	Mukasurat/ Page : 35 Drpd / of :8

TAJUK/TITLE: SAFE MACHINE OPERATION

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the Safe Machine Operation. The person who is competent in this CA shall be able to abide to Safety and Health Acts, and Factories and Machinery Acts and regulations, hence reducing exposure to hazards and risks as well as incidence of work-related injury and illness.

NO. KOD /CODE NO.	Z-009-1:2015–M04/P(3/5)	MukaSurat / Page : Drpd/ of: 9
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PENERANGAN/INFORMATION:

1. Introduction

Potential hazards of operating machines and equipment are numerous. Some of the most obvious recognized hazards are from machine motion. Hazardous motion is characteristic of the point-of-operation of the machine, but can also be found in other areas such as behind, to the side, or above the machine.

- 1.1 Rotating motion of collars, couplings, cams, clutches, flywheels, shaft-ends, set screws, spindles, etc., can be dangerous by gripping clothing or forcing arms/hands or other body parts into dangerous positions. Rotating parts can also create nip points when two adjacent moving parts are in close proximity (e.g., two cogs, two rolling bars, chain and sprocket, etc.); or a rotating part is in close proximity to a fixed point.
- 1.2 Reciprocating machine motions are also hazardous. A worker may be injured by back-and forth or up-and-down motion when struck by or caught between moving and stationary parts (e.g., saw blades, knives, etc.).
- 1.3 Transverse machine motion (movement in a straight, continuous line) is another recognized hazard because a worker may be struck or caught in a pinch or shear point by moving parts.

Why is safeguarding important? It is impossible to predict what people will do around powered machinery, no matter what their level of experience or training. Effective safeguarding protects against both human error and lapses in judgment, as well as accidental contact with moving parts.

For instant view of the impact, while adjusting a conveyor belt, a millwright is drawn into the unguarded tail drive of a belt conveyor and suffers fatal crushing injuries. Another case, a young worker feeding sheet rubber into a guillotine shear loses both hands when the machine cycles unexpectedly. And a case of a lumber piler cleaning up around a lumber sorting table (green chain) is strangled when his loose clothing is caught on an exposed keyway at the end of a slowly rotating shaft. These are three examples of serious injury or death resulting from unguarded or inadequately safeguarded machinery and

NO. KOD /CODE NO.	Z-009-1:2015–M04/P(3/5)	MukaSurat / Page : Drpd/ of: 9
--------------------------	--------------------------------	---

equipment which can happen in any workplace that uses powered equipment. Each year, physical contact with machines and powered equipment account for a significant number of life-altering injuries, including amputations and disfigurement, and may also result in the death of a worker. Most of these accidents can be prevented by effective safeguarding, lockout procedures, and supervision and training.

2. Fencing Machinery

- 2.1 Every dangerous part of every driven machine, prime mover and transmission machinery shall be securely fenced in accordance with these Regulations: Provided that where the Chief Inspector is satisfied that there is available and suitable for use in connection with machinery any type or description of safety device, he may direct such type or description of device shall be provided for use in connection with such machinery.
- 2.2 Where the Chief Inspector is satisfied that owing to special circumstances the compliance with any of these Regulations is unnecessary or impracticable he may by certificate in writing, which he may in his discretion revoke, exempt the whole or any part of any machinery from the provisions of such regulation.
- 2.3 Opening in fence.
No opening in any fence shall be of such dimensions that any person can touch the moving parts of the machinery enclosed by the fence, provided that guard-rails may be installed where the clearance between the nearest exposed moving part of the machinery and the guard-rail exceeds twelve inches but does not exceed twenty inches.
- 2.4 Point of operation.
The point of operation of any machine shall be effectively fenced.
- 2.5 Starting and stopping appliance.
 - i. Every machine intended to be driven by mechanical power shall be provided with an efficient starting and stopping appliance the control of which shall be in such a position as to be readily and conveniently operated by the person operating the machine in his normal working position.

NO. KOD /CODE NO.	Z-009-1:2015–M04/P(3/5)	MukaSurat / Page : Drpd/ of: 9
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- ii. Where pedals are used to actuate machines or parts of machines, an automatic locking device shall be attached to the pedal or an inverted U-shape guard shall be fastened to the floor over the treadle leaving sufficient clearance for the foot of the operator between the treadle and guard.
- iii. No person shall start any machinery unless and until every precaution has been taken to ensure that no other person or persons will be injured or be rendered liable to injury or be placed in a hazardous position by the operation of such machinery.

2.6 Electrical apparatus.

All electrical equipments and installations shall be of such construction and so installed and maintained as to prevent fire hazard and danger from contact with moving parts and live parts. Such electrical equipments and installations shall conform to all the requirements prescribed in any written law relating to electrical equipments and installations and shall have been approved by the authorities appointed by the said law.

Rotating motion can be dangerous; even smooth, slowly rotating shafts can grip clothing and force an arm or hand into a dangerous position. Injuries due to contact with rotating parts can be severe (see **figure 1**).

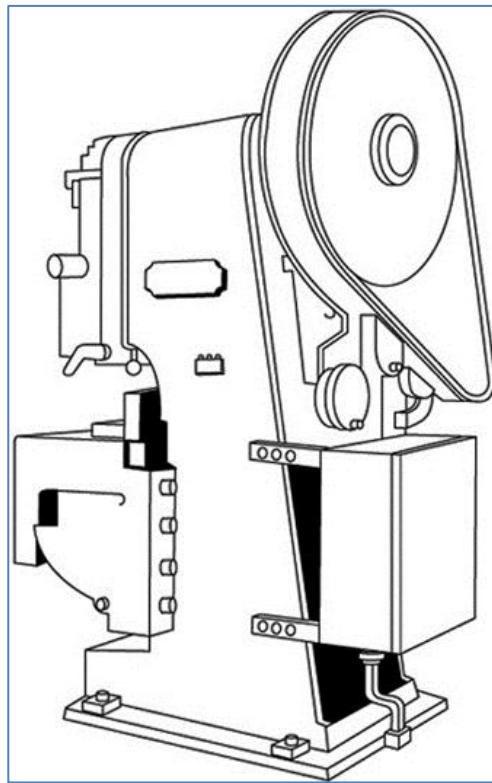


Figure 1: Mechanical punch press

Collars, couplings, cams, clutches, flywheels, shaft ends, spindles and horizontal or vertical shafting are some examples of common rotating mechanisms which may be hazardous. There is added danger when bolts, nicks, abrasions and projecting keys or set screws are exposed on rotating parts on machinery, as shown in **figure 2**.

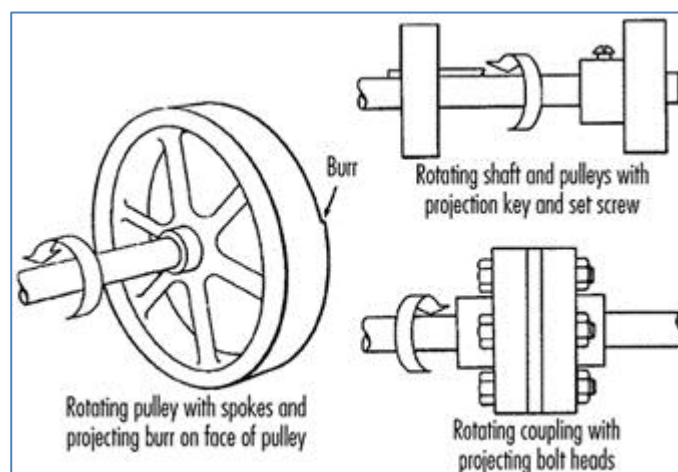


Figure 2: Examples of hazardous projections on rotating parts

In-running nip points are created by rotating parts on machinery. There are three main types of in-running nip points:

Parts with parallel axes can rotate in opposite directions. These parts may be in contact (thereby producing a nip point) or in close proximity to each other, in which case the stock fed between the rolls produces the nip points. This danger is common on machinery with intermeshing gears, rolling mills and calenders, as shown in **figure 3**.

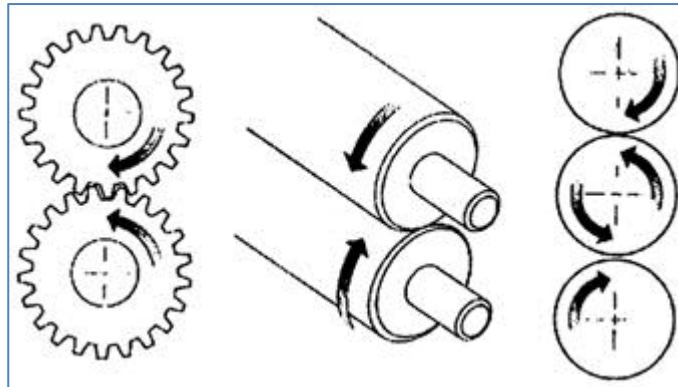


Figure 3: Common nip points on rotating parts

Another type of nip point is created between rotating and tangentially moving parts, such as the point of contact between a power transmission belt and its pulley, a chain and a sprocket, or a rack and pinion, as shown in **figure 4**.

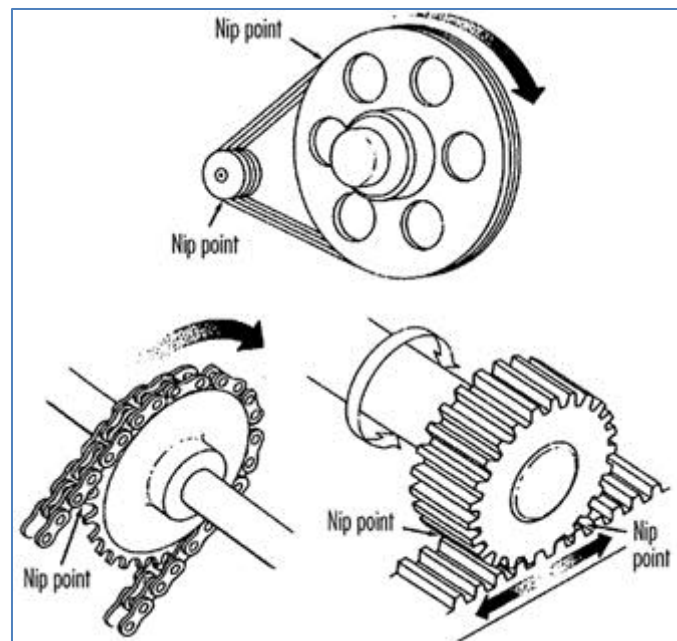


Figure 4: Nip points between rotating elements and parts with longitudinal motions

Nip points can also occur between rotating and fixed parts which create a shearing, crushing or abrading action. Examples include hand-wheels or flywheels with spokes, screw conveyors or the periphery of an abrasive wheel and an incorrectly adjusted work rest, as shown in **figure 5**.

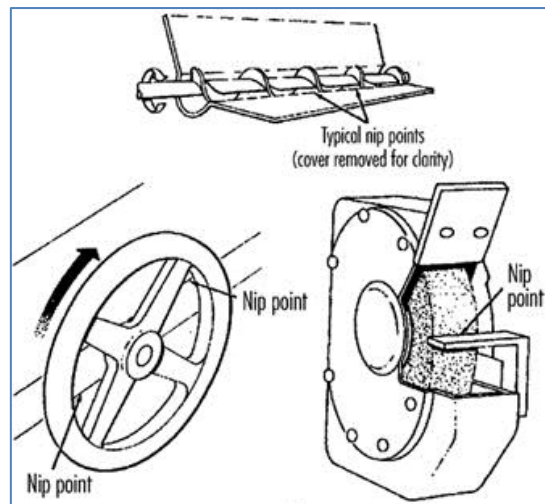


Figure 5: Nip points between rotating machine components

Transverse motion (movement in a straight, continuous line) creates a hazard because a worker may be struck or caught in a pinch or shear point by a moving part. An example of transverse motion is shown in **figure 6**.

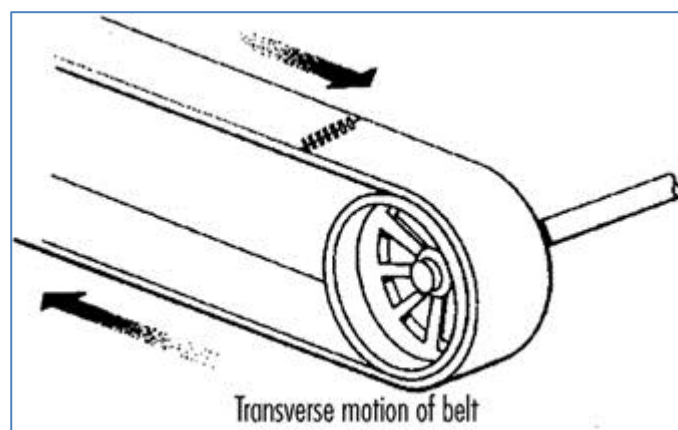


Figure 6: Example Of Transverse Motion

Cutting action involves rotating, reciprocating or transverse motion. Cutting action creates hazards at the point of operation where finger, head and arm injuries can occur and where flying chips or scrap material can strike the eyes or face. Typical examples of machines

with cutting hazards include band saws, circular saws, boring or drilling machines, turning machines (lathes) and milling machines. (See **figure 7**).

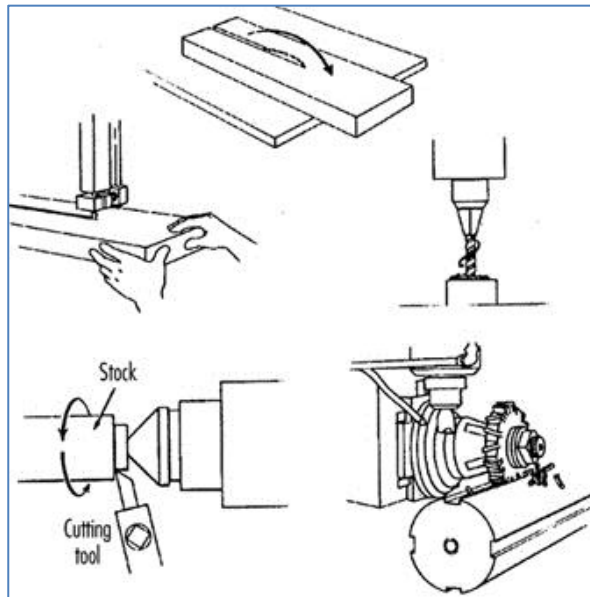


Figure 7: Examples of cutting hazards

SOALAN/QUESTION :

1. Why machine guarding is important for any operation of machinery?

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2. Name three types of machine guarding?

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3. What type of guarding for a cutting machine?

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NO. KOD /CODE NO.	Z-009-1:2015–M04/P(3/5)	MukaSurat / Page : Drpd/ of: 9
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